

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

April 2013

EE-201 CONTROL ENGINEERING

Date: 22.04.2013, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q-1 (a) Define the following terms 03
- State , Control system , Feedback Path
- (b) Explain servomechanism system for position control system by giving appropriate example. 04
- Q-2 (a) Obtain the transfer function using Mason's Gain formulae for the signal flow graph shown in figure I. 07

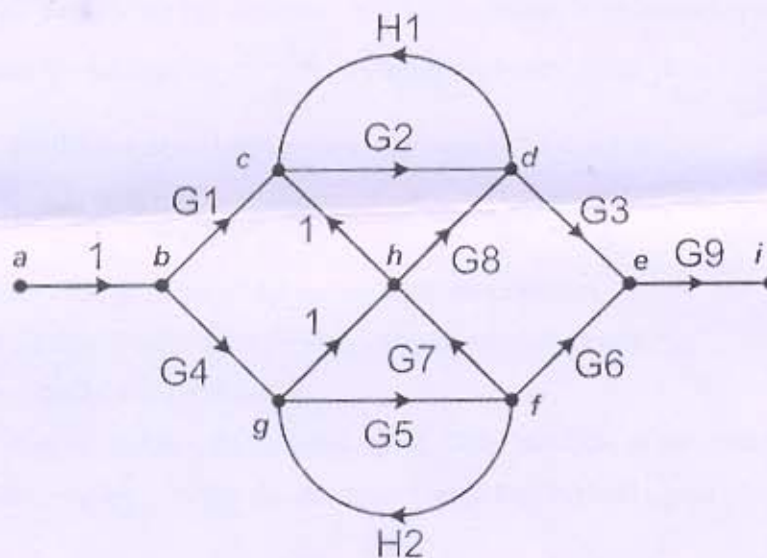


Figure – I

- (b) Derive the transfer function for Gear train mechanism. 07

OR

- Q-2 (a) Obtain the state space representation of the following differential equation and draw the state diagram for the same. 04

$$\frac{d^4 y(t)}{dt^4} + 5 \frac{d^3 y(t)}{dt^3} + 3 \frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} - 2 y(t) = u(t)$$

(b) Comment on Controllability & Observability of the system described by

04

$$\dot{x} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u(t)$$

$$y = [1 \quad 0 \quad 2] x(t)$$

(c) Obtain Transfer Function using Block Diagram Reduction method for the block diagram shown in Figure II.

06

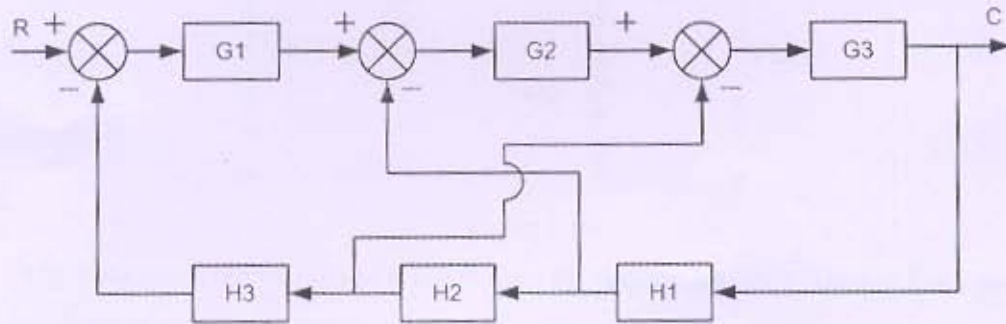


Figure- II

Q-3 (a) Prove that by introducing PD controller in 2nd order closed loop control system the damping ratio is increased by $\frac{\omega_n K_d}{2}$. (Assume necessary parameters.)

05

(b) For a unity feedback system having open loop transfer function as

05

$$G(s) = \frac{K(s+2)}{s^2(s^2+7s+12)}$$

Determine (i) Type & Order of the system (ii) Error constants

(iii) Steady state error for step, ramp & parabolic input.

(c) (i) Explain the following terms.

03

- Peak Overshoot, Ramp Input signal, Order and type of the system.

(ii) Give the relation between the step signal input and impulse signal input.

01

OR

Q-3 (a) Derive the expression of c(t) for unit step input for underdamped case for second order system.

07

(b) List out different types of standard input signals. Give the expression of input of each signal in Time domain and Laplace domain.

05

(c) The poles of the second order system is given by -1-2j, -1+2j. Locate it on a s-plane

02

& find out value of ζ .

SECTION – II

Q-4(a) Obtain the transfer function from the given circuit shown in figure III.

05

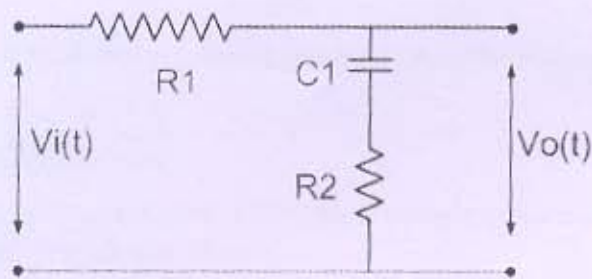


Figure III

(b) Why state space representation is used? Give the relation between Transfer function and State space representation.

02

Q-5 (a) A unity feedback control system has an open loop transfer function

07

$$G(s) = \frac{K}{s(s+3)(s^2+2s+2)}$$

Sketch the root locus of the system for positive values of K on Graph paper.

(b) The characteristics equation of the system is given by $s^4+20s^3+15s^2+2s+K=0$

07

- (i) Determine the range of K for the system is being stable using Routh's criteria.
- (ii) Find out value of K for becoming the system marginally stable.
- (iii) Find Frequency of oscillations.

OR

Q-5 (a) Sketch the root locus with the system with

07

$$G(s)H(s) = \frac{K}{s(s+1)(s+3)}$$

Sketch the root locus on graph paper.

(b) The characteristics equation of the system is given by $s^6+4s^5+3s^4-16s^2-64s+48=0$
Comment on stability.

07

Q-6 (a) Sketch the Bode plot on the Semilog graph paper for the transfer function

07

$$G(s) = \frac{KS^2}{s(s+0.2s)(1+0.02s)}$$

- (b) For a feedback control system ,

07

$$G(s) H(s) = \frac{40}{(s+4)(s^2+2s+2)}.$$
 Comment on stability using Nyquist Plot.

OR

- Q-6 (a) Sketch the nature of Bode plot on semilog graph paper for the system having

07

$$G(s) H(s) = \frac{80}{s(s+2)(s+20)}.$$

- (b) What do you mean by compensation? List types of compensation techniques used in basic control system. Explain any of them.

07
